

COLOMBO RATMALANA, Sri Lanka

WMO: 434670

Lat: 6.822N

Lon: 79.886E

Elev: 7

StdP: 101.24

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.8	23.4	17.7	12.7	27.1	18.7	13.6	26.9	7.1	27.3	6.3	28.2	1.6	50	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	4.9	33.2	26.6	32.8	26.4	32.4	26.3	27.7	31.6	27.5	31.3	27.2	31.0	3.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.2	30.0	26.3	21.8	29.9	26.1	21.5	29.8	88.4	31.9	87.1	31.6	86.0	31.4	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.2	4.7	DB	19.6	35.9	1.4	2.3	18.6	37.6	17.8	38.9	17.0	40.2	16.0	41.8	(4)
(5)			WB	18.3	28.4	1.5	0.4	17.2	28.6	16.3	28.9	15.5	29.1	14.3	29.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	27.2	27.8	28.8	29.1	29.3	28.8	28.7	28.4	28.3	28.1	27.6	27.3
	DBStd	1.18	1.05	0.94	0.72	1.05	1.12	0.97	1.01	0.97	1.09	0.92	0.82	0.97
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6673	534	498	582	572	599	565	579	571	549	560	527	535
	CDD18.3	3631	276	265	323	322	341	315	321	313	299	302	277	277
	CDH23.3	41849	2764	2886	3997	4037	4388	3889	3894	3669	3455	3376	2792	2701
	CDH26.7	16031	921	1052	1732	1841	1989	1549	1487	1283	1213	1210	911	843
Wind	WSAvg	2.1	2.4	2.0	1.7	1.6	2.3	2.5	2.5	2.5	2.2	1.7	1.7	2.1
Precipitation	PrecAvg	2469	74	76	121	243	358	192	136	104	227	369	336	171
	PrecMax	3920	211	220	357	618	752	602	482	271	551	871	971	477
	PrecMin	1774	0	0	1	71	48	30	11	3	20	95	59	19
	PrecStd	479	56	61	74	131	185	102	107	72	132	168	165	108
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	33.2	33.5	33.9	33.3	32.2	32.2	31.8	32.1	32.2	32.2	32.4
		MCWB	23.9	24.9	26.0	27.4	27.4	26.9	26.7	26.5	26.4	26.0	25.2	24.1
	2%	DB	31.9	32.2	32.9	33.2	32.8	31.7	31.5	31.2	31.6	31.7	31.6	31.5
		MCWB	24.1	25.1	26.1	27.1	27.2	26.8	26.4	26.2	26.2	26.0	25.3	24.5
	5%	DB	31.2	31.6	32.5	32.8	32.4	31.3	31.0	30.8	31.1	31.2	31.2	30.9
		MCWB	24.3	25.1	26.0	26.9	27.1	26.7	26.2	26.1	26.0	25.9	25.4	24.8
	10%	DB	30.5	31.0	32.0	32.4	31.9	30.9	30.5	30.3	30.6	30.8	30.7	30.3
		MCWB	24.2	25.1	25.9	26.8	26.9	26.6	26.1	25.9	25.9	25.8	25.4	24.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.7	27.2	28.1	28.2	27.7	27.2	27.0	27.1	26.9	26.7	26.5
		MCDB	30.2	30.5	31.6	32.5	31.9	30.8	31.5	31.2	30.9	30.3	30.2	29.7
	2%	WB	25.7	26.3	27.0	27.7	27.7	27.2	26.7	26.6	26.7	26.6	26.2	26.1
		MCDB	29.6	30.2	31.5	32.1	31.4	30.5	30.4	30.3	30.3	30.0	29.9	29.4
	5%	WB	25.4	26.0	26.6	27.4	27.5	27.1	26.6	26.4	26.4	26.3	26.0	25.8
		MCDB	29.2	30.0	31.1	31.7	31.2	30.4	30.1	29.8	29.9	29.8	29.7	29.1
	10%	WB	25.1	25.7	26.4	27.1	27.2	26.8	26.3	26.2	26.2	26.1	25.7	25.5
		MCDB	28.8	29.5	30.8	31.4	30.8	30.0	29.7	29.5	29.5	29.7	29.2	28.7
Mean Daily Temperature Range	5% DB	MDBR	7.1	6.9	6.9	6.7	4.9	4.1	3.8	3.9	4.4	5.6	6.2	6.2
		MCDBR	7.7	7.4	7.2	7.0	5.5	4.5	4.1	4.5	5.1	6.0	6.9	7.0
		MCWBR	3.1	2.8	2.5	2.6	2.0	1.6	1.5	1.5	1.6	1.9	2.3	2.5
	5% WB	MCDBR	6.6	6.8	6.9	6.7	5.2	4.2	4.1	4.4	4.7	5.6	6.5	6.4
		MCWBR	2.9	2.9	2.5	2.8	2.1	1.7	1.7	1.7	1.8	2.1	2.3	2.6
Clear-Sky Solar Irradiance	taub	0.438	0.454	0.497	0.500	0.516	0.498	0.514	0.494	0.463	0.470	0.469	0.455	
	taud	2.364	2.302	2.196	2.214	2.157	2.222	2.150	2.226	2.330	2.306	2.304	2.330	
	Ebn at Noon	879	874	836	820	789	793	784	815	852	848	843	854	
	Edh at Noon	125	136	153	148	152	140	152	144	132	134	132	128	
All-Sky Solar Radiation	RadAvg	5.33	5.97	6.28	5.93	5.31	5.28	5.41	5.51	5.56	5.32	4.72	4.74	
	RadStd	0.37	0.32	0.29	0.30	0.40	0.31	0.25	0.28	0.39	0.33	0.34	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	+0.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page